Yuwei Du

Master's candidate in the Department of Electronic Engineering at Tsinghua University, specializing in LLM agents, spatial intelligence, urban science and spatiotemporal data mining. My research focuses on developing agentic frameworks for human behavior modeling, spatiotemporal reasoning, and automated scientific discovery in transportation. Published papers in top-tier venues including KDD, NAACL, NeurIPS, and CoNEXT. Advised by Professor Jian Yuan and Professor Yong Li at Tsinghua University.

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EDUCATION

College of Computer Science and Electronic Engineering, Tsinghua University *Master Degree*, Sep. 2023 -- Jun. 2026

- Main Courses: Matrix analysis and application, big data analysis, data mining, machine learning, statistical learning methods.
- Cumulative GPA: 91/100.

School of Electronic Information and Electrical Engineering, Beijing Jiaotong University *Bachelor Degree*, Sep. 2018 -- Jun. 2022

- Main Courses: Advanced Mathematics, Principles of Communications, Fundamentals of Information Theory
- Cumulative GPA: 89/100.

SELECTED RESEARCH EXPERIENCE

An Autonomous AI Scientist for Automated Traffic Problem-Solving with Simulation-Validated Optimization Nov. 2025 -- Mar 2026

- Engineered a full-stack toolchain via the Model Context Protocol (MCP), providing holistic support across the research lifecycle, including multi-dimensional sensing, multi-level control, and autonomous network/demand editing.
- Curated a multi-grained benchmark covering a full spectrum of traffic tasks and conducted instruction-tuning on domain-specific models to enhance spatiotemporal reasoning and task execution.
- Architected an end-to-end "AI Scientist" framework that automates the complete traffic research pipeline—from autonomous hypothesis formulation and simulation-validated experimentation to iterative reflection and optimization of governance strategies.

Automated City Traffic Simulation Powered by Agent Framework Jun. 2025 -- Oct 2025

- Designed a hierarchical multi-agent framework that decomposes natural language instructions into executable simulation workflows, enabling adaptive task orchestration in complex urban traffic scenarios.
- Developed a goal-oriented reinforcement learning strategy with memory-augmented reasoning, allowing LLM agents to autonomously optimize traffic condition in traffic simulator.
- Implemented an MCP-based interface for secure, iterative interaction between LLMs and simulation tools, establishing a closed-loop “perceive-decide-act-reflect” pipeline that surpasses rigid, predefined simulation workflow.

Automated Trajectory Modeling via Collaborative Learning of Large and Small Models Dec. 2024 -- Jun. 2025

- Proposed an end-to-end automated trajectory modeling agent framework that collaborates with single-task small models, comprising task identification → model selection → automatic optimization → deployment/training → result analysis
- Established standardized interfaces between agents and specialized models by integrating trajectory modeling architectures with heterogeneous structures, creating a unified execution environment, and designing harmonized workflows for processing heterogeneous data (including GPS, road networks, and check-in records).
- Incorporated a performance metric-driven adaptive optimization mechanism that enables dynamic data augmentation, parameter tuning, and joint optimization.

Agent Framework for Trajectory Generation Based on City Knowledge Foundation Model May. 2024--Nov. 2024

- Proposed an agent framework capable of generating urban mobility trajectories for arbitrary user profiles.
- Enhanced the spatial comprehension capabilities of city knowledge foundation model through fine-tuned with fine-grained geographic knowledge corpora, effectively mitigating hallucinations in trajectory generation tasks.
- Proposed a three-step generation method: (1) extracting template mobility patterns and synthesizing individualized ones, (2) mapping mobility patterns to real urban spaces using an enhanced city knowledge foundation model, and (3) employing a DPO alignment strategy to enhance the spatiotemporal continuity of generated trajectories.

Generalizable Cross-Region Trajectory Prediction Using an Agent Framework Dec. 2023--Mar. 2024

- Proposed an agent framework capable of next-location prediction in data-sparse regions.
- Propose a method leveraging large models' world knowledge to model urban structure's influence on mobility patterns, and a multi-agent mobility pattern sharing scheme that employs graph networks to model the joint effects of social relationships and collective transition regularities on individual mobility patterns.
- Enhanced prediction accuracy by integrating individual mobility patterns, collective transition regularities, and geographic knowledge of target city.

Boosting Urban Spatial Awareness and Reasoning Capabilities in Large Language Models Dec. 2023--Mar. 2024

- Construct a diversified instruction-tuning dataset to infuse urban knowledge into large language models and enhance their spatial reasoning capabilities.
- Proposed a two-stage hybrid fine-tuning approach that combines urban instruction data with open-source general-purpose instruction data. Introduced a self-weighted adjustment mechanism for fine-tuning, which dynamically guides the model to focus on high-quality, well-matched training samples during optimization.
- Developed a multidimensional urban scenario and geospatial task evaluation benchmark to assess trained models' urban knowledge cognition capabilities.

SELECTED PUBLICATIONS

[1]Jie Feng, Yuwei Du, Jie Zhao, et al. "Agentmove: Predicting Human Mobility Anywhere Using Large Language Model Based Agentic Framework," in Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2025), 2025. (CCF B)

[2]**Yuwei Du**, Jie Feng, Jie Zhao, et al. "TrajAgent: An Agent Framework for Unified Trajectory Modelling," in *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025. (CCF A)

[3]Jie Feng*, Tianhui Liu*, **Yuwei Du**, et al. "CityGPT: Empowering Urban Spatial Cognition of Large Language Models," in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025)*, 2025. (CCF A)
[4]Jie Feng, Jun Zhang, Junbo Yan, **Yuwei Du**, et al. "CityBench: Evaluating the Capabilities of Large Language Model as World Model," in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025)*, 2025. (CCF A)
[5]Ruoxuan Yang, **Yuwei Du**, Hongyang Du. "Mapping the Wizards' Path: A Systematic Review of Wizard-of-Oz in HCI with Implications for the LLM Era," in *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI 2026)*, 2026. (CCF A)
[6]Yibo Ma, Tong Li, **Yuwei Du**, et al. "Sustainable Connections: Exploring Energy Efficiency in 5G Networks," in *Proceedings of the 20th International Conference on Emerging Networking Experiments and Technologies (CoNEXT 2024)*, 2024. (CCF B)
[7]**Yuwei Du**, Jie Zhao, Jun Zhang, et al. "TrafficScientist: Towards an Autonomous AI Scientist for Automated Traffic Problem-Solving with Simulation-Validated Optimization," Under Review at The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026, Industry Track), 2026. (CCF A)
[8]**Yuwei Du**, Jie Feng, Jian Yuan, et al. "GAMS: Geospatial Knowledge Enhanced Agentic Framework for Urban Human Mobility Behavior Simulation," Under Review at ACL Rolling Review (January 2026 Cycle), 2026. (CCF A)
[9]**Yuwei Du**, Jun Zhang, Jie Feng, et al. "TrafficSimAgent: A Hierarchical Agent Framework for Autonomous Traffic Simulation with MCP Control." Under Review at the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026, Research Track), 2026. (CCF A)
[10]Shiteng Cao, Xiaochong Lan, **Yuwei Du**, et al. "Enhancing Local Life Service Recommendation with Agentic Reasoning in Large Language Models." Under Review at the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026, Research Track), 2026. (CCF A)
[11]Shiyuan Zhang, **Yuwei Du**, Ruoxuan Yang, et al. "U-SCOPE: Multi-Agent Adaptive User Modeling for Personalized Recommendations in Mobile AI Applications," *Under Review at Advances in Neural Information Processing Systems (NeurIPS 2026)*, 2026. (CCF A)